Activity Title: Fuzzify and Defuzzify (Understanding Fuzzy Classification)

Objective:
To apply the concepts of fuzzification and defuzzification by defining membership functions, calculating fuzzy values, and converting them back into crisp values.

Instructions (Individual Task)

1. **Scenario:**
   You are designing a simple fuzzy controller for an air conditioner.
   The input variable is **"Room Temperature (°C)"** and the output variable is **"Fan Speed."**
2. **Step 1 – Define Membership Functions (Fuzzification):**
   Define fuzzy sets for temperature:
   - *Cold*
   - *Warm*
   - *Hot*

   Example range: 15°C – 35°C
   Use **triangular or trapezoidal** membership functions (students can sketch or describe in words).

3. **Step 2 – Assign Membership Values:**
   For a room temperature of **28°C**, calculate or estimate the membership degree for each fuzzy set (Cold, Warm, Hot).
   (e.g., $\mu Cold = 0.0$, $\mu Warm = 0.4$, $\mu Hot = 0.6$)
4. **Step 3 – Apply a Simple Rule Base:**
   - IF Temperature is *Cold*, THEN Fan Speed is *Low*
   - IF Temperature is *Warm*, THEN Fan Speed is *Medium*
   - IF Temperature is *Hot*, THEN Fan Speed is *High*

   Combine the fuzzy outputs for the given input using **aggregation**.

5. **Step 4 – Defuzzification:**
   Use any one method (student's choice) —
   - Max-membership
   - Centroid
   - Mean of Maximum

   Compute or describe the **crisp output** (Fan Speed) value.

Expected Deliverables

Each student submits:

A short write-up (1 page max) including:

- ✓ Defined membership functions (with rough sketch)
- ✓ Membership degree calculations
- ✓ Chosen defuzzification method and final output
- ✓ 2–3 sentences reflecting on how fuzziness helps improve real-world decision-making.